

Hao Cao 曹浩

Associate Professor, Department of Earth, Planetary, and Space Sciences
University of California, Los Angeles

UCLA EPSS, 4859 Slichter Hall, Los Angeles, CA 90095 · hcao@epss.ucla.edu · 310-794-9467 · haocao.epss.ucla.edu

RESEARCH INTERESTS

Planetary geophysics and space environment · planetary magnetism and dynamos · gravitational and electromagnetic sounding of planetary interiors · planetary exploration, including space missions and space-based magnetometry.

HONORS AND AWARDS

2025	Sloan Research Fellowship in Physics , Alfred P. Sloan Foundation
Mar. 2022	Certificate of Distinction in Teaching , Harvard University
Aug. 2018	NASA Group Achievement Award , Cassini MAG Team, NASA
Sep. 2012	Eugene B. Waggoner Scholarship , UCLA
May 2012	NASA Earth and Space Science Fellowship , NASA

PARTICIPATION IN SPACE MISSIONS

Feb. 2024-present	Deputy Instrument Lead/Co-I , NASA TRACERS MAG Team
Jan. 2016-present	Co-Investigator , ESA JUICE J-MAG Team
Aug. 2014-present	Member , Juno Science Team
Mar. 2015-Sep. 2017	Cassini Participating Scientist , Cassini Mission
Jun. 2014-Sep. 2017	Member , Cassini MAG Team

SCIENTIFIC INSTRUMENT DEVELOPMENT

Apr. 2024-present	Principal Investigator , NASA DALI Project L-MAG (Lunar Magnetometer)
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ACADEMIC APPOINTMENTS

Jul. 2026-present	Associate Professor , University of California, Los Angeles
Jul. 2022-Jun. 2026	Assistant Professor , University of California, Los Angeles
Sep. 2017-Jun. 2022	Research Associate , Harvard University
Aug. 2014-Sep. 2017	Postdoctoral Scholar , California Institute of Technology

EDUCATION

Jun. 2014	Ph.D. , Geophysics and Space Physics, UCLA
Jun. 2011	M.S. , Geophysics and Space Physics, UCLA
Jun. 2009	B.S. , Geophysics and Space Physics, University of Science and Technology of China

VISITING POSITIONS

Sep.-Dec. 2022	Simons Fellow , Isaac Newton Institute for Mathematical Sciences, Cambridge University
Jul. 2022-Jul. 2024	Associate , Harvard University
Feb. 2018-present	Academic Visitor in Physics , Imperial College London
Sep. 2017-Sep. 2020	Visitor in Planetary Science , California Institute of Technology

SELECTED PUBLICATIONS

1. Agiwal, O., **Cao, H.**, Hsu, H.W., et al. (2024). Current Events at Saturn: Ring-planet Electromagnetic Coupling. *Planetary Science Journal* 5, 134. doi:10.3847/PSJ/ad4343
2. Bloxham, J., **Cao, H.**, Stevenson, D.J., et al. (2024). A rapidly time-varying equatorial jet in Jupiter's deep interior. *Nature* 627, 64-66. doi:10.1038/s41586-024-07046-3
3. **Cao, H.**, Bloxham, J., Park, R.S., et al. (2023). Strong resemblance between surface and deep zonal winds inside Jupiter revealed by high-degree gravity moments. *The Astrophysical Journal* 959, 78. doi:10.3847/1538-4357/ad0cbb
4. **Cao, H.**, Dougherty, M.K., Hunt, G.J., et al. (2023). Saturn's magnetic field at unprecedented detail achieved by Cassini's close encounters. In *Saturn after the Cassini Grand Finale*, Cambridge University Press. arXiv:2301.02756
5. Yadav, R.K., **Cao, H.**, Bloxham, J. (2022). A global simulation of the dynamo, zonal jets, and vortices on Saturn. *The Astrophysical Journal* 940, 185. doi:10.3847/1538-4357/ac9d94
6. Agiwal, O., **Cao, H.**, Cowley, S.W.H., et al. (2021). Constraining the temporal variability of neutral winds in Saturn's low-latitude ionosphere using magnetic field measurements. *JGR: Planets* 126, e2020JE006578. doi:10.1029/2020JE006578
7. **Cao, H.**, Dougherty, M.K., Hunt, G.J., et al. (2020). The landscape of Saturn's internal magnetic field from the Cassini Grand Finale. *Icarus* 344, 113541. doi:10.1016/j.icarus.2019.113541

Full publication list: haocao.epss.ucla.edu/publications/

NATIONAL AND INTERNATIONAL COMMITTEES

- Apr. 2025-present **Member**, JUICE-Clipper Steering Committee, ESA JUICE Project and NASA Europa Clipper Project
- Mar. 2023-May 2026 **UCLA Representative** to the Council of Institutions (COI), Universities Space Research Association (USRA)

PROFESSIONAL SERVICE

Conference roles. SOC Co-chair, NASA TRACERS Science Team Meeting (Sep. 2024); LOC Co-chair, Russell Symposium (Sep. 2024); SOC Member, DPS55/EPSC, San Antonio, TX (2023); Session Convener/Chair, AGU Fall Meeting (2018, 2023-2025), DPS55/EPSC (2023), DPS48/EPSC11, Pasadena (2016).

Journal referee. Nature; Nature Astronomy; Nature Geosciences; Space Science Reviews; Nature Communications; Proceedings of the National Academy of Sciences; The Astrophysical Journal; Icarus; Journal of Geophysical Research - Planets; Geophysical Research Letters; Earth and Planetary Science Letters; Progress in Earth and Planetary Science; Monthly Notices of the Royal Astronomical Society; Scientific Reports; Advances in Space Research; Atmosphere; Earth and Planetary Physics; Frontiers in Physics.

INVITED SEMINARS AND COLLOQUIA

Caltech, Planetary Science Seminar (Apr. 2026) · Imperial College London, Physics Department Seminar (Nov. 2025) · UC Berkeley, CIPS Seminar (Apr. 2025) · Caltech, Planetary Science Seminar (Feb. 2023) · Jet Propulsion Laboratory, ICE Seminar (Jan. 2023) · University of Washington Seattle, ESS Colloquium (Apr. 2022) · UCLA, Space Physics Seminar (Feb. 2022) · ETH Zurich, Department of Earth Sciences ERC Seminar (Jan. 2021) · UC Berkeley, Astronomy/EPS Joint Colloquium (Nov. 2019) · Boston University, Center for Space Physics Seminar (Mar. 2019) · Georgia Institute of Technology, EAS Seminar (Feb. 2019) · Imperial College London, Space & Atmospheric Physics Seminar (Mar. 2018) · University of Leicester, Astro Seminar (Mar. 2018) · University of Cambridge, DAMTP Astrophysics Seminar (Feb. 2018) · MIT, EAPS Planetary Lunch (Feb. 2018) · Cornell, Astronomy and Space Sciences Special Seminar (Feb. 2018) · Cornell, Astronomy and Space Sciences Colloquium (Feb. 2018) · Harvard, EPS GSPD Seminar (Feb. 2018) · Carnegie DTM, Astro Seminar (Jan. 2018) · Imperial College London, Space & Atmospheric Physics Seminar (Nov. 2017) · UT Austin, Institute of Geophysics Seminar (Nov. 2016) · Jet Propulsion Laboratory, Planetary Science Seminar (Sep. 2016) · UC Santa Cruz, Whole Earth Seminar (Feb. 2016) · UCLA, Planet Lunch (Jan. 2016) · High Altitude Observatory (HAO), Boulder, Colloquium (Apr. 2015) · Caltech, Yuk Lunch Seminar (Mar. 2015) · UCLA, iPLEX Lunch Seminar (May 2013).